

Volitional vs. Forced: A Field Study of Just-in-Time Interventions for Digital Attention Recovery

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Abstract

This study investigates two mechanisms for the intervention of mobile attention, self-commitment and forced lockout, in an in-the-wild setting. Drawing on Locus of Control theory, a Just-in-Time Adaptive Intervention (JITAI) was implemented, triggered by elevated app-switching behavior. Although both interventions were implemented after distraction from the task, the self-commitment condition was associated with a smaller decrease in task willingness, indicating a greater potential to support motivation recovery. These findings underscore the promise of low-friction, autonomy-respecting designs for future personalized attention-aware systems.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; *HCI design and evaluation methods*.

Keywords

Attention Management, Just-in-Time Adaptive Intervention (JITAI), Volitional Intervention, Locus of Control, Digital Boredom, Autonomy in HCI, In-the-Wild Study

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1 Introduction

In digital environments, the constant bombardment of information and sensory input is reshaping the structure of attention in modern individuals. Users often switch rapidly between different content streams to avoid sustained engagement. This phenomenon has led us to propose a new concept: **digital boredom**.

According to the MAC model [3, 13], boredom results from a joint failure of attentional engagement and perceived meaning. While traditional boredom arises from a lack of stimulation, digital boredom emerges precisely in contexts of overstimulation. In such states, users appear hyperactive on the surface yet remain psychologically inert—unable to focus or truly rest. Recent findings [12] further suggest that frequent switching behavior not only fails to alleviate this mental discomfort but instead exacerbates fatigue, attentional fragmentation, and a loss of meaning, forming a self-reinforcing cycle of attentional collapse.

To address this gap, we developed a lightweight **JITAI** (Just-in-Time Adaptive Intervention) system that detects frequent app-switching behavior and proactively sends self-commitment prompts, encouraging users to voluntarily return to their primary task.

Our findings demonstrate that autonomy-supportive attention interventions offer a compelling alternative to control-based mechanisms in mobile interaction design.

2 Related Work

2.1 Attention Management through External Control

Existing works show that current approaches to digital attention regulation rely mostly on **externally imposed controls**, such as screen-time limits and lockout systems [1, 7]. While these interventions can reduce overuse in the short term, they frequently compromise user autonomy and engender frustration [1, 8]. As a result, these restrictions can trigger **psychological reactance**, where users seek to bypass or resist perceived control, or even behavioural backlash such as increased engagement with restricted

apps [7, 8]. Softer friction-based interventions have also been proposed as ways to disrupt automatic mobile interactions without breaking the overall experience flow. For example, Cox et al. propose “**microboundaries**” such as delayed loading and confirmation dialogues [1]. However, the challenge in this approach lies in balancing context and frequency subtly enough to disrupt habitual use without triggering circumvention [1, 10].

2.2 Motivation-Based and Volitional Interventions

Studies suggest that **volitional interventions** which focus on users’ internal motivation and agency could support self-regulated attention recovery [2, 4, 6]. Examples include **implementation intentions** linking situational cues to intended behaviour and **pre-commitment prompts** for proactive boundary setting [4, 6]. These interventions centre on self-initiation and are grounded in the **Locus of Control** [11] and **Self-Determination** theories, which emphasize autonomy and competence as key determinants of motivation and sustaining attention [2, 13]. While Hiniker et al.’s “MyTime” app demonstrated the potential of volitional interventions on smartphone use in naturalistic settings, deployment of such approaches in **mobile attention regulation**, especially in **in-the-wild environments**, remains limited [6, 8].

2.3 Boredom, Hyperattention, and Digital Overload

Recent works suggest that **digital boredom** represents a gap in intervention design. Boredom arises from both **failed attentional engagement** and **lack of meaning**, as proposed by Eastwood et al. in the MAC model (Meaning and Attention Components), later formalized by Westgate and Wilson [3, 13]. Tam and Inzlicht found that rapid exposure to short-form video content and content switching creates a feedback loop that not only exacerbates this boredom and mental fatigue, but also reduces attention and user satisfaction [12]. Han outlines this societal shift from deep, contemplative attention to **hyperattention**, arguing that continuous digital stimulation deprives individuals of the capacity for “deep boredom” necessary for creativity and rest [5]. To navigate the nuances in addressing digital boredom and sustainable attention recovery, more interventions need to balance contextual salience and autonomy [5, 13]. Nahum-Shani et al. formalize one such approach in the Just-in-Time Adaptive Intervention (JITAI) framework, which emphasizes intervening at moments of vulnerability or opportunity [9]. **Our study therefore utilizes this principle in comparing volitional self-commitment against forced lockout mechanisms in everyday attention recovery scenarios.**

3 System Architecture and Intervention Design

We developed an application called the **Digital Boredom Adaptive Intervention System (DBAIS)**. This application is available on Android only. Based on Nahum Shani et al.’s Just-in-Time Adaptive Intervention (JITAI) framework, DBAIS detects and responds to users’ boredom states without breaking their usage flow [5]. DBAIS consists of the following features: its **operation flow** (Figure 1), **two types of interventions** (Section 3.2), and an **intelligent measurement mechanism** (Section 3.3).

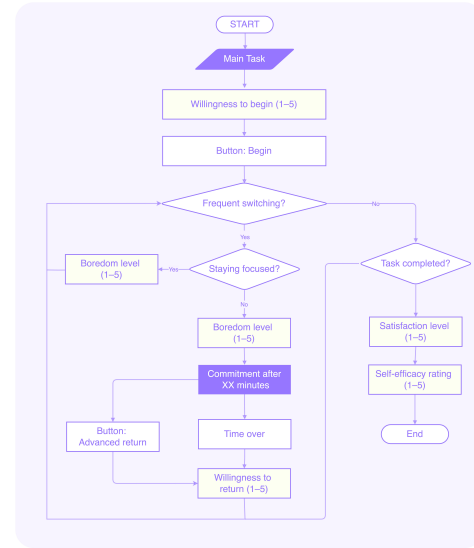


Figure 1: System Architecture Flowchart (Volitional Commitment)

3.1 System Operation Flow

Users launch DBAIS and enter two inputs: their current task and initial willingness to engage in it. The system then monitors app usage in the background. If five or more app switches occur within four minutes, a pop-up appears to assess the user’s attention and boredom levels. If the user reports being distracted, an intervention is triggered, using one of two modes: volitional commitment (Figure 1) or forced intervention, as set in the system. After completion of the task, users rate their satisfaction and self-efficacy, which are used to evaluate the effectiveness of the intervention (Figure 2).

3.2 Intervention Mechanism Design

Two attention intervention mechanisms were developed and are switchable via system settings.

3.2.1 Volitional Commitment Mechanism. When users’ attention drifts, DBAIS prompts them to set a delay before returning to the task. Users commit to starting the task after a set time (e.g., 15 minutes), and a countdown is displayed in the notification bar, encouraging an early return. When the set time ends, a pop-up asks users about their willingness to resume the task. Users can delay their commitment multiple times, simulating procrastination. After completing the task, users rate their satisfaction and self-efficacy.

3.2.2 Forced Intervention Mechanism. This mechanism reshapes users’ attention by restricting access to non-task-related apps until the task is completed. Users first set a whitelist of allowed apps. If the intervention is triggered, DBAIS provides a 5-minute buffer with a countdown. Once it ends, non-whitelisted apps are blocked. Users then record their willingness to return, followed by a satisfaction and self-efficacy rating upon task completion.

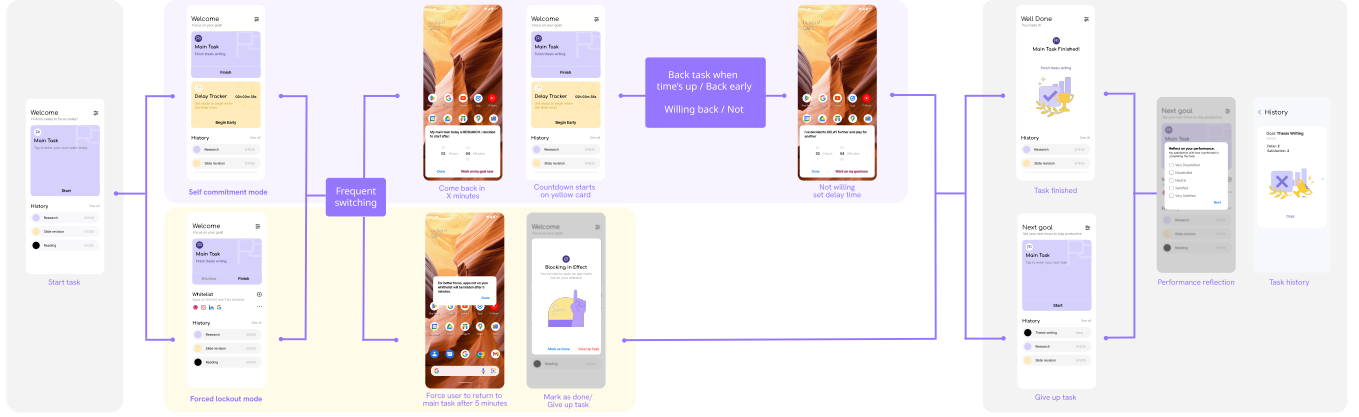


Figure 2: UI Design(Self-Commitment+Forced Lockout)

3.3 Intelligent Behavior Measurement

All experimental data is stored locally on users' devices and then forwarded to us. We use GPT-4 to process certain experimental metrics (e.g., determining if users return to the main task or become distracted based on app-switching patterns). These metrics are then reviewed and discussed by our team to reach a consensus on the data-driven indicators of task return and distraction.

4 Experimental Procedures

We conducted an in-wild A / B field experiment to compare the efficacy of voluntary commitment and forced intervention mechanisms in addressing user digital boredom and returning to their main task. Our experiment consisted of two stages: a preliminary experiment used to determine the trigger frequency threshold to identify digital boredom, and a main experiment.

4.1 Preliminary Experiment

We used the initial module of DBAIS to monitor app-switching behavior and determine the optimal trigger frequency for our JITAI. This preliminary test was conducted with three female participants ($M = 24.67$, $SD = 1.53$) on the same device. During the test, participants were asked to self-rate their current level of boredom on a 5-point Likert scale each time a pop-up was triggered under different app-switching frequency conditions. A total of 172 boredom ratings were recorded (Table 1).

We tested six frequency conditions, including three key conditions—5 switches per 3, 4, and 5 minutes—with 42 ratings collected for each. Participants rated their current boredom level on a 5-point Likert scale (1 = not bored at all, 5 = extremely bored). Based on the average boredom score ($Avg = 4.31$), rating consistency ($SD = 0.869$), and natural trigger frequency, we selected 5 switches per 4 minutes as the optimal intervention threshold for the main experiment.

4.2 Main Experiment

We retained this boredom rating mechanism in the main experiment to verify the applicability of this threshold in real-world mobile usage scenarios. The experiment was conducted over two days per participant, within the period of May 21–25, 2025. Before

Table 1: Switching Condition Data

Cond.(min/switch)	Size	Avg. Score	SD	Freq. (act./min)
5 / 3	16	3.625	1.204	0.6
5 / 4	24	3.917	0.881	0.8
5 / 5	42	3.833	0.881	1
4 / 5	42	4.310	0.869	1.25
3 / 5	42	4.381	0.936	1.67
3 / 6	6	4.333	0.816	2

each round, the system prompted participants to enter their main task (Task X) and rate their willingness to start from 1 to 5 points. The system then passively monitored app usage in the background. When five or more app switches were detected within four minutes, the intervention was triggered naturally in the usage flow. The volitional commitment mechanism employed a softer approach via a self-determined notification countdown, while the forced intervention mechanism enforced a lockout system from non-whitelisted apps. On task completion, we recorded ratings for satisfaction and self-efficacy.

We recruited a total of 14 voluntary participants (7 male, 7 female), aged 18–26 ($M = 22.64$, $SD = 2.31$), including students and working adults. Each participant completed one round each for the two mechanisms, resulting in 28 valid rounds. DBAIS recorded complete app usage trajectories, subjective scale ratings, and behavioural resumption data.

This study adopts a within-subject design, where each participant experiences both of these attention intervention mechanisms at different times. We randomly balanced the order of interventions as an A/B test to control for order effects. To best preserve natural behavior, all experiments were conducted without guided operations or behavioral disguises in participants' regular daily usage contexts and on their own Android devices. To ensure compatibility across devices, we adapted and debugged UI presentation, notification styles, and permission logic for consistent intervention logic across different devices (Table 2).

Table 2: Participant Details

ID	G	Age	Device	Ver.	Order
P01	F	19	VIVO x90pro	13	AB
P02	F	22	Xiaomi 13	13	AB
P03	M	21	Honor Magic5	15	AB
P04	F	18	VIVO x100s	14	AB
P05	M	21	VIVO IQOO 10 Pro	14	AB
P06	M	24	Honor MAGIC5	15	AB
P07	M	24	Xiaomi 13	14	AB
P08	F	23	Xiaomi 12s Pro	13	BA
P09	M	22	Xiaomi 11 Pro	14	BA
P10	F	25	Xiaomi 15 Ultra	15	BA
P11	M	24	Xiaomi Civi4 Pro	15	BA
P12	F	26	Realme gt5Pro	15	BA
P13	F	25	Galaxy s22	13	BA
P14	M	23	VIVO x100	15	BA

4.3 Data Collection and Processing

We designed a structured user data collection framework across four categories to capture multidimensional data during our experiment process. These were used to evaluate the efficacy of the two intervention mechanisms. These four categories comprise:

4.3.1 Contextual Inputs. Active inputs from users during the experiment. For example, main task statement and custom delay times (not exceeding 3 hours).

4.3.2 Self-Reported Measures. Users complete three scales on willingness, satisfaction and self-efficacy using five-point Likert scales (1 to 5). These are done at task setup, post-intervention, and task completion.

4.3.3 Objective Behavioural Logs. DBAIS automatically records core usage behaviours, including number of intervention triggers, app usage paths, and durations.

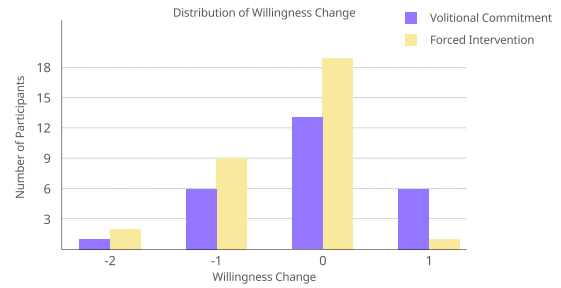
4.3.4 Derived Measures via LLM. GPT-4 assisted in labeling whether each event involved a **task return**—defined as resuming task-related app use within approximately 5 minutes after the intervention—and a **distraction relapse**—defined as using non-task apps after re-engagement or before task completion, within 25 minutes post-intervention. All labels were cross-validated by the team and used to compute return and relapse rates for both groups.

This layered approach enables a robust and interpretable evaluation of intervention outcomes.

5 Results and Discussion

We recorded a total of 56 intervention events in an in-the-wild setting, including 26 from the volitional commitment group and 30 from the forced lockout group. Each participant experienced both intervention types, and all analyses were conducted at the level of individual intervention events ($N = 56$), not participants. Task willingness was assessed using a 5-point Likert scale (1 = very unwilling, 5 = very willing) before and after each intervention.

Across both conditions, users generally exhibited a decline in task willingness after resuming from distraction. However, the average


Figure 3: Distribution of Willingness Change

decrease was smaller in the volitional group ($\Delta M = -0.07$) than in the forced group ($\Delta M = -0.30$), suggesting a potential advantage of self-initiated commitment in sustaining task motivation.

To explore directional effects, we examined the number of intervention events in which task willingness increased versus did not increase. In the volitional group, 6 out of 26 events showed an increase, compared to only 1 out of 30 in the forced group (Figure 3). A 2×2 analysis revealed a significant difference via Fisher's exact test ($p = 0.048$, $OR = 7.91$), indicating that volitional commitment significantly increased the likelihood of motivational improvement. Pearson's chi-square test approached significance ($\chi^2 = 2.93$, $p = 0.087$), further supporting this trend.

No significant differences were found in self-reported satisfaction or self-efficacy, suggesting that intervention type primarily affects short-term motivational states rather than affective or cognitive appraisals. We also compared distraction and task re-engagement rates. The volitional group had a higher distraction rate (0.79) but a similar re-engagement rate (0.36) compared to the forced group (0.63 and 0.37, respectively).

Qualitative inspection revealed that some participants in the forced group whitelisted non-task apps, effectively bypassing the lockout mechanism. While forced interventions reduced overt app-switching, they also prompted strategic avoidance behaviors.

Although our preliminary experiment suggested a correlation between switching frequency and boredom, this pattern did not hold in the main study. Observed individual differences—such as high-frequency switching without boredom—diluted the correlation. Nonetheless, the frequency threshold functioned as a reliable intervention trigger.

Taken together, this study presents preliminary evidence that volitional commitment mechanisms may better preserve task motivation in real-world mobile contexts, compared to forced lockout interventions. Grounded in Locus of Control theory, our results highlight the motivational benefits of autonomy-supportive designs and the broader promise of applying psychological theory to attention interventions. Future work will expand this direction through larger-scale validation, task-type control, and refinement of boredom detection signals.

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